



Effectiveness of one minute per level intercostal nerve cryoablation for postoperative analgesia after surgical correction of pectus excavatum

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ABSTRACT

Purpose: Intraoperative intercostal nerve cryoablation has emerged as a promising modality for postoperative analgesia following Surgical Correction of Pectus Excavatum (SCOPE). Most centers use two-minute cryoablation per level, although data from histologic and adult studies suggest the effectiveness of one-minute freezes. We aimed to describe our center's experience with one minute per level cryoablation.

Methods: A retrospective single institution review of patients undergoing SCOPE was performed to compare patients pre- and post-intercostal nerve cryoablation implementation. Cryoablation was performed as one minute for each of the thoracic intercostal nerves T3–T7. Multivariable regression analyses were conducted to compare the outcomes and cost between pre- and post-implementation groups.

Results: During the study period, 198 patients underwent SCOPE with one Nuss bar, receiving either intraoperative intercostal nerve cryoablation (Cryo, $n = 100$) or preoperative thoracic paravertebral catheters (NoCryo, $n = 98$). Surgical time was on average 9 min longer for the Cryo group ($p < 0.01$). Median length of stay for the Cryo group was 3 days shorter compared to the NoCryo group ($p < 0.01$). The Cryo group had a 19-fold and 5.6-fold reduction in average inpatient and total postoperative opioid usage, respectively ($p < 0.01$). Total hospital costs were significantly lower in the Cryo group ($p < 0.01$). Overall complication rate was not statistically significant different between the two groups.

Conclusions: Intraoperative one minute per level cryoablation is a potent approach to postoperative analgesia for SCOPE patients that led to a shorter hospital length of stay, lower hospital costs, and decreased opioid use compared to conventional analgesia at our institution. Pediatric surgeons performing correction of chest wall deformities should consider offering this technique.

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1. Introduction

Pectus excavatum (PE) is the most common chest wall deformity in children and adolescents with a prevalence of 1–8 per 1000 worldwide [1,2]. It is characterized by a depression of the sternum eliciting functional and psychological stress on the affected individuals causing significant morbidity and affecting their quality of life [3–6]. The minimally invasive Nuss procedure has

been the gold standard for Surgical Correction of Pectus Excavatum (SCOPE) since its introduction in the late 1990s. It involves the placement of one or more metal bars behind the sternum to reshape the chest wall [7–10]. However, despite its advantages, this procedure is still associated with severe prolonged postoperative pain, mandating a significant hospital stay [11,12].

Intercostal nerve cryoablation (Cryo) involves applying the cryoablation probe to intercostal bundles at temperatures of -60° to -70° °C and causing nerve alterations associated with temporary paresthesia of the affected nerves [13–29]. Cryo has been gaining popularity as an alternative method of pain management for SCOPE due to its association with prolonged analgesic effect, lower opioid utilization, and earlier discharge [13–27]. Cryo has also been associated with lower overall cost despite the use of additional equipment and the added operative time to apply Cryo to 4 or 5 nerves bilaterally [13,23]. These outcomes have revolutionized the care for PE. However, while most currently published reports

Abbreviations: PE, Pectus Excavatum; SCOPE, Surgical Correction of Pectus Excavatum; Cryo, Intercostal nerve cryoablation; PCA, Patient-Controlled Anesthesia; MME, Oral Morphine Milligram Equivalents; POD, Postoperative Day; LOS, Length of Stay; PACU, Post anesthesia care unit; ERAS, Enhanced recovery after surgery; Sch, Scheduled; PRN, As needed.

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describe applying Cryo for two minutes at each intercostal nerve level, there is a lack of evidence surrounding the optimal duration for each application of Cryo. Histologic studies have shown equivalent changes in the nerves in terms of axonal degeneration, accumulation of edema, and capillary stasis with intact endoneurium with one- or two-minute freezes [30,31]. Clinical reports from as early as 1980 have reported the effectiveness of one-minute freeze times in thoracotomy patients [30,32]. However, one-minute freeze has not yet been described for minimally invasive techniques or in children. We aimed to describe our institutional experience with one minute per level Cryo during SCOPE and compare their outcomes and hospital costs with SCOPE patients from the pre-Cryo implementation period.

2. Methods

2.1. Study cohort and demographic variables

Our institution started performing one minute per level Cryo for SCOPE patients in July 2020. Therefore, we obtained Institutional Review Board approval to evaluate the effectiveness of Cryo implementation and compare it with pre-implementation SCOPE outcomes. We conducted a retrospective cohort review of patients undergoing SCOPE at our institution between July 2018 and September 2021. We excluded patients prior to July 2018, since a standardized perioperative patient care protocol was implemented for SCOPE patients at our institution at that time. Only patients who underwent SCOPE with one bar without concurrent operations were included. For each patient, we collected age, gender (male/female), race (white/non-white), ethnicity (Hispanic/non-Hispanic), insurance status (public/non-public), weight, BMI, and Haller index.

2.2. Intercostal nerve cryoablation technique

Since its implementation at our institution in July 2020, Cryo has been performed using the CryoICE probe (©AtriCure, Inc., Mason, OH). A lateral chest incision is made in preparation for bar insertion. A 30° thoracoscope is inserted at the medial part of the incision and the CryoICE probe is inserted at the lateral part of the same incision with a separate pleural entry. We do not use double-lumen endotracheal tube or single lung ventilation. The pleural space is insufflated with carbon dioxide (CO₂) and tidal volumes are decreased. With low lung volumes and mild changes in the tilt of the operative table we have been able to navigate the probe to an appropriate location at the proximal aspect of the intercostal bundle, just lateral to the musculo-cartilaginous junction. Each nerve is then frozen at −60 °C for one minute. This process is repeated on intercostal nerves 3–7 bilaterally. With the assistance of thoracoscopic visualization, we also administer percutaneous local anesthetic in each nerve space to aid in short term pain control until the analgesic effects of Cryo begin to take effect. The rest of the operation proceeds in the usual fashion.

2.3. Perioperative protocol (Fig. 1)

Prior to the implementation of Cryo, our institution developed a protocolized perioperative approach for patients undergoing SCOPE to decrease length of stay and improve pain control and patient satisfaction. Starting with the preoperative appointment, patients and families received education and counseling on the expected pain and hospital course, as well as anxiety management strategies. Preoperatively on the day of surgery, patients received one dose of Pregabalin, before undergoing bilateral paravertebral nerve blocks with indwelling infusion catheter placement. At the

end of the operation, patients received a one-time dose of intravenous (IV) Acetaminophen. On arrival to the post anesthesia care unit (PACU), a Ketamine infusion in addition to a Hydromorphone patient-controlled anesthesia (PCA) with continuous infusion rate were started. Afterwards, patients were started on scheduled IV Diazepam, Ketorolac, oral Acetaminophen, and Pregabalin on postoperative day (POD) 0. Over the next two days, IV and paravertebral infusions were stopped, and patients transitioned to oral (PO) alternatives. Patients were discharged home once their pain was well controlled on an oral pain medication regimen.

After implementation of Cryo, preoperative education was adjusted based on anticipated shorter hospital stay. Patients continued to receive one dose of Pregabalin preoperatively and a dose of IV Acetaminophen post operatively. No Paravertebral blocks, PCA pumps, or Ketamine infusion were started. All patients were started on oral Acetaminophen, Ibuprofen, Diazepam, and Oxycodone on POD0 compared to POD3 prior to Cryo implementation. Patients were discharged home after their pain was well controlled on an oral pain medication regimen.

At each follow-up visit, our patients pre- and post-Cryo implementation were queried about neuropathic pain-related symptoms, those who still complained of pain after a week of NSAID treatment were prescribed Gabapentin for management of neuropathic pain. However, Gabapentin prescriptions were not a routine part of our protocol, so patients receiving gabapentin in the office were treated as having a complication of neuropathic pain.

2.4. Primary and secondary outcomes

Opioid utilization reported as oral morphine milligram equivalents (MME) was our primary outcome of interest. Secondary outcomes included operating room time, surgical cut-to-close time, non-surgical operating room time, length of stay, total hospital stay cost, and the rate of associated complications. Complications were identified as bar displacement, surgical site infection, pneumonia, pleural effusion, need for chest tube postoperatively, neuropathic pain, any return to the operating room (OR), or 30-day readmission. We also collected data on complaints of rash in the area of distribution of the nerves affected by Cryo and the number of phone calls in the first 30 days postoperatively.

2.5. Statistical analysis

Analysis was performed comparing Cryo and non-Cryo (NoCryo) groups. Pearson Chi-squared analysis was used to compare categorical variables between groups. Continuous variables were reported as median with Interquartile range (IQR) and were compared using T-test for parametric variables and Wilcoxon rank-sum (Mann-Whitney) test for non-parametric variables. Kaplan-Meier survival curve was used for graphical representation of length of stay. Adjusting for demographics and intraoperative differences, multivariable linear regression analyses were conducted to compare operating room times, opioid utilization, and cost, while multivariable logistic regression was used to compare the overall complication rate between the two groups. All analyses were performed using STATA (StataCorp. 2021. Stata Statistical Software: Release 17. College Station, TX: StataCorp LLC.). Alpha level of 0.05 was considered for statistical significance.

3. Results

3.1. Cohort characteristics

During our retrospective review period, a total of 198 patients underwent SCOPE with one Nuss bar, of which 100 underwent

Table 1
Preoperative patient characteristics.

	Cryo	NoCryo	p-value
Total, N (%)	100 (50.51%)	98 (49.49%)	
Age (years), median [IQR]	15 [14–16]	15 [14–17]	0.57
Male, N (%)	84 (84%)	84 (85.71%)	0.74
White, N (%)	77 (77%)	64 (65.31%)	0.07
Hispanic, N (%)	19 (19%)	28 (28.57%)	0.24
Non-public insurance, N (%)	72 (72%)	76 (77.55%)	0.37
Weight (Kg), median [IQR]	56.3 [50.4 – 63.2]	55.3 [49 – 61.7]	0.42
BMI, median [IQR]	18.27 [17.05 – 20.19]	18.4 [17.2 – 20.5]	0.77
Haller index, median [IQR]	3.9 [3.4 – 4.6]	4.1 [3.4 – 5]	0.37

Table 2
Unadjusted intraoperative descriptive characteristics.

	Cryo	NoCryo	p-value
Surgery cut-to-close (minute), median [IQR]	59.5 [54–69]	52 [40–65]	<0.01
Total operating room time (minute), median [IQR]	119 [107–128.5]	114.5 [104–133]	0.43
Patients needing rib/partial sternal resection, N (%)	4 (4%)	2 (2.04%)	0.42

Table 3
Unadjusted postoperative length of stay, opioid utilization, and cost.

	Cryo Median [IQR]	NoCryo Median [IQR]	p-value
LOS, days	1 [1–2]	4 [3–4]	<0.01
Inpatient MME, mg	20.7 [11.5–36.8]	409.1 [313.6–534.6]	<0.01
Inpatient MME/kg, mg	0.4 [0.2–0.7]	7.5 [5.8–9.2]	<0.01
Total postoperative MME, mg	109 [93.3–160.4]	628.2 [488.4–732.4]	<0.01
Total postoperative MME/kg, mg	2 [1.7–2.6]	11.1 [8.6–13.2]	<0.01
Total hospital cost, \$	14,072.7 [13,126.6–16,348.4]	21,021.8 [19,380.1–23,695.3]	<0.01

MME: Oral Morphine Milligram Equivalent, POD: Postoperative Day, LOS: Length of Stay, Total includes inpatient usage and outpatient prescription.

Cryo. There were no statistically significant differences in preoperative patient characteristics between the two groups (Table 1).

3.2. Operative characteristics

Unadjusted intraoperative characteristics for each group are shown in Table 2. On multivariate linear regression and after adjusting for age, gender, race, ethnicity, insurance, BMI, Haller index, and the need for rib or partial sternal resection, the Cryo group was associated with a 9-minute longer surgery cut-to-close time (9, 95%CI [4–14]). However, there was no significant difference in total operating room time between the two groups ($p = 0.7$).

3.3. Postoperative outcomes

Table 3 shows unadjusted postoperative LOS, opioid utilization, and cost. 90% ($N = 90$) of the Cryo group patients were discharged home by POD2 compared to none from the NoCryo group (Fig. 2). The Cryo group had a 19-fold and 5.6-fold reduction in average inpatient and total postoperative opioid usage per kilogram, respectively ($p < 0.01$) (Fig. 3).

Using multivariable linear regression, the Cryo group was associated with lower overall opioid utilization, with an adjusted average of 399 fewer inpatient MMEs (-398.6 , 95%CI [-435.6 – -361.7]) and 502 fewer total postoperative MMEs (-502.4 , 95%CI [-549.2 – -455.6]) utilized per patient.

3.4. Complications

Table 4 describes the unadjusted incidence of all complications in addition to the total number of phone calls within 30 days and complaints of rash. Indications for phone call included discomfort,

Table 4
Unadjusted postoperative complication rates.

	Cryo (%)	NoCryo (%)	p-value
Neuropathic Pain	3 (3%)	9 (9.2%)	0.07
Surgical Site Infection	5 (5%)	7 (7.1%)	0.49
Bar Displacement*	0 (0%)	5 (5.1%)	0.02
Pneumothorax requiring chest tube	6 (6%)	3 (3.1%)	0.32
Pulmonary Effusion	1 (1%)	0 (0%)	0.32
Pneumonia	1 (1%)	2 (2%)	0.55
Return to OR*	0 (0%)	6 (6.1%)	0.01
30- day Readmission	7 (7%)	5 (5.1%)	0.58
Any complication	15 (15%)	25 (25.5%)	0.07
Rash	9 (9%)	4 (4.1%)	0.16
30-day number of Calls to Surgery Clinic	50 (50%)	53 (54.1%)	0.57

* $p < 0.05$.

rash, and questions about medication intake. On multivariable logistic regression analysis, Cryo and NoCryo group did not show a significant difference in odds of overall complication rate (OR 0.5, 95%CI [0.24–1.02]).

4. Discussion

Our study is among the largest studies of Cryo use for perioperative analgesia during SCOPE to date and the only report to our knowledge of one minute per level Cryo times in this patient population. This study demonstrates that the implementation of Cryo was associated with shorter hospital length of stay and lower opioid utilization in our patients. These findings are consistent with previous reports in the literature and highlight the effectiveness of Cryo even at shorter application times [13–27,33,34].

With opioid-related causes now the 3rd leading cause of death among Americans under 19 years of age, our findings of a 19-

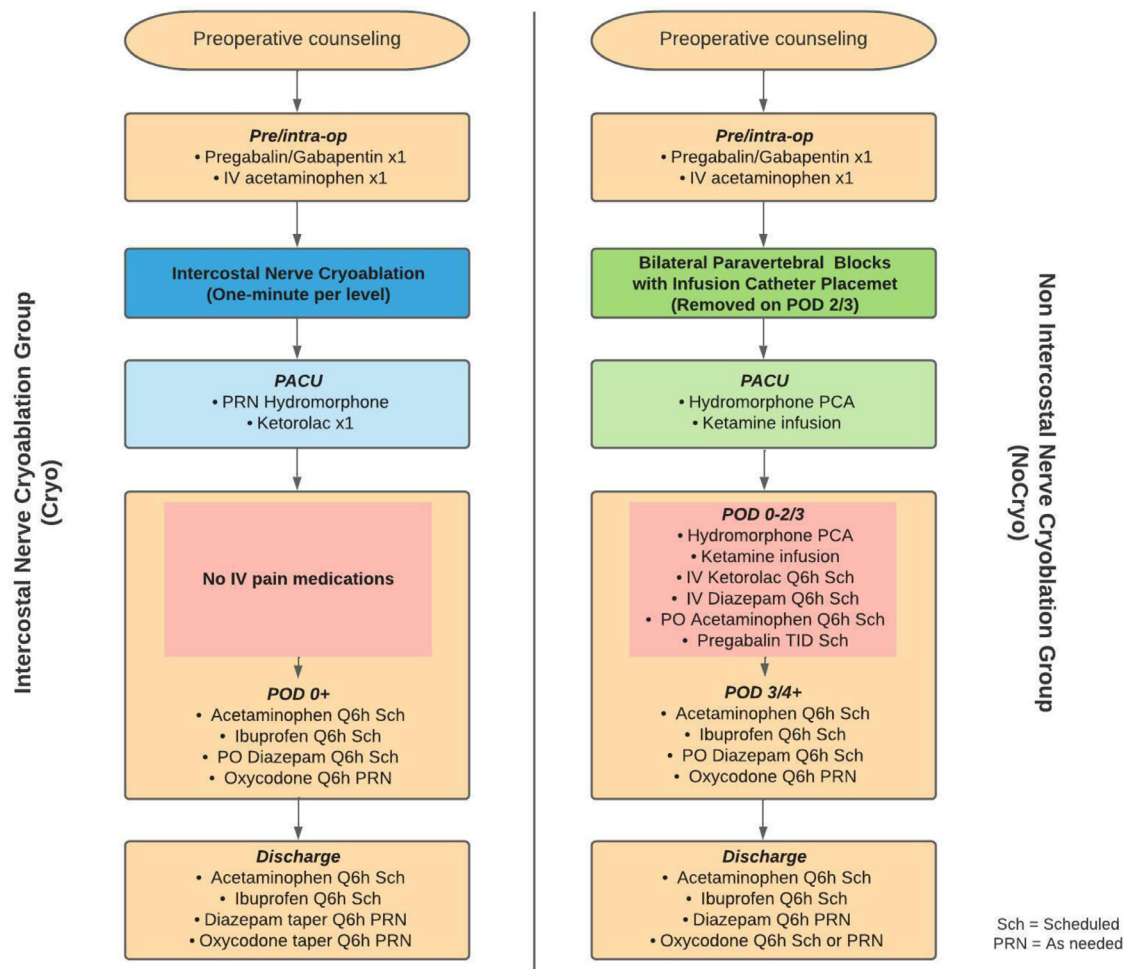


Fig. 1. Perioperative protocol.

fold decrease in inpatient opioid utilization after SCOPE demonstrates the value of Cryo as an effective tool for decreasing opioid exposure and availability in this patient population, [1,3,9,22,35]. The impact of Cryo on opioid utilization extends across the outpatient setting, where our results also show a 5.6-fold decrease in total opioid utilization in the ambulatory postoperative period, with fewer than half the number of pills prescribed prior to implementation of Cryo per our protocol. Our findings of decreased opioid utilization echo most of the previously reported literature [13–15,18–21,23,25,36,37].

In addition to decreasing opioid utilization amidst the opioid epidemic, Cryo implementation was also associated with a significant decrease in hospital length of stay for SCOPE patients in our cohort from a median of 4 days to 1 day. This marked reduction converts most patients' hospital stays from inpatient to observation status, dramatically reducing inpatient resource utilization for these patients. This finding comes at a time when there is a significantly high demand inpatient healthcare, especially in the setting of the COVID-19 pandemic. This reduction in hospital days allowed for more beds and staff to be available for additional patients in need of care. This is especially important in hospitals at which NoCryo SCOPE patients may have needed ICU admissions postoperatively for monitoring [9]. Our findings of decreased length of stay are concordant with other recent reports showing discharge on POD1 for most Cryo patients and an improvement over initial institutional Cryo reports that described a length of stay longer than 2 days [13–27,36,37].

Cryo replaced paravertebral block infusion catheters in our SCOPE perioperative protocol, which is an important consideration for the generalizability of our results. Catheter placement was linked to a longer length of stay for observation while the catheters were in place, a duration of at least 2 days in our pre-Cryo protocol. Plus, other than decreasing the length of stay needed because of those paravertebral catheters, Cryo implementation obviates the need for the additional preoperative procedure for catheter placement while the patient is usually awake.

Unlike previous reports describing Cryo for SCOPE using the manufacturer general setting of two minutes, we used one minute per level freezing time. Implementing Cryo for SCOPE with these one-minute freeze times provided our patients high quality analgesia without significantly changing our overall OR times compared to pre-Cryo times, despite paravertebral blocks being performed outside the OR. By performing Cryo, our findings point to an improved overall utilization and efficiency of the hospital's perioperative services for patients undergoing SCOPE, mainly by decreasing the preoperative needs of these patients. While surgical time is about 9 min longer using Cryo, this increase in time is outweighed by the time that would have been needed for bilateral placement of paravertebral infusion catheters, which is closer to 30 min in our cohort, thereby reducing overall burden on anesthesia providers and the preoperative staff and improving the perioperative experience for patients.

The aforementioned benefits and the decrease of opioid use during the whole postoperative period stem from the physiology

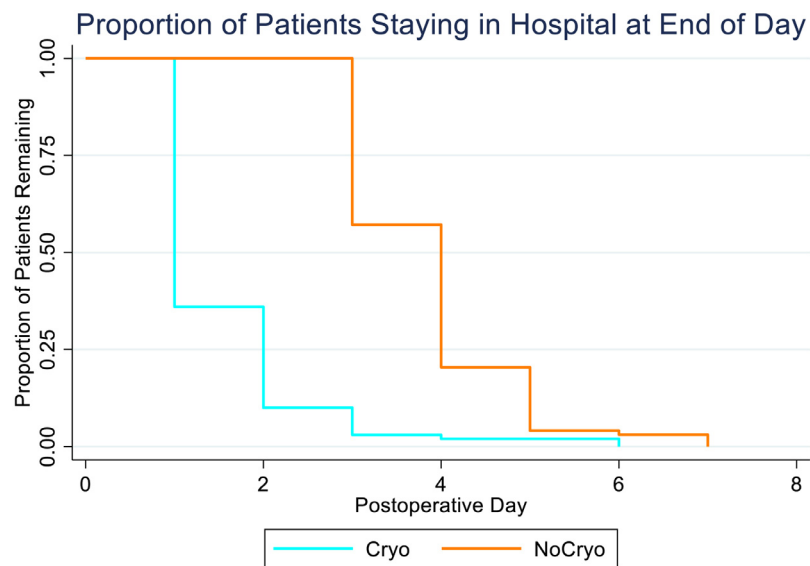


Fig. 2. Kaplan-Meier curve showing proportion of patients remaining the hospital at the end of each postoperative day.

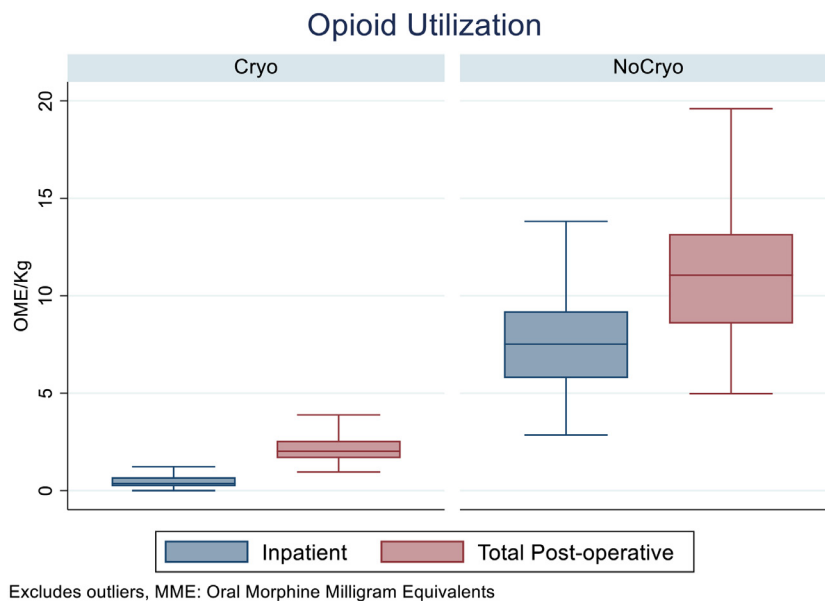


Fig. 3. Inpatient and total postoperative opioid utilization comparison between the two groups.

of cryoablation. Cryoablation applied to nerves at a temperature of -60°C is associated with immediate nerve swelling and delayed Mullerian degeneration while leaving the basal membranes intact [30,31]. These changes are linked to temporary paresthesia of the affected nerve until nerve regeneration is complete. Histological evidence shows complete regeneration by 30 days, though restoration of complete function may lag up to a year and the time to recovery was proportionally related to the freeze time [30,31]. Considering that the optimal duration of application of Cryo remains unknown along with our findings of effective pain control, we hypothesize that shorter, one minute per level, freeze times could be associated with fewer long term side effects and complications related to Cryo without compromising its efficacy. This is the aim of an ongoing prospective study.

Our study did not find a significant difference in the overall complication rates between the Cryo and NoCryo groups. Furthermore, Cryo was significantly associated with fewer occurrences of bar displacement and all cause return to the operating room. These

results highlight the overall safety profile of Cryo implementation. Prior studies demonstrated some concern over increased bar displacement rates with utilization of Cryo, but we did not experience a single episode of bar displacement in the 100 patients described in this study [17,18,22,25]. Some authors hypothesized that superior pain control with Cryo may have led patients to an early sense of security and foregoing of activity restrictions leading to bar displacement [18,25,38]. However, other studies did not demonstrate such results or did not report an increase in bar displacement [20]. At our institution, we include focused pre- and postoperative counseling on activity restrictions that may have been associated with our low bar displacement rate. In addition, the overall low number of bar displacement in our cohort may be related to the high volume of SCOPE procedures performed at our institution, with high surgeon-volume found to be associated with a lower odds of complications in a recent study [39].

The high complication rate in our study is owed in part to inclusion of unplanned outpatient Gabapentin prescriptions as a

complication in an attempt to capture neuropathic pain. In addition, about 9% of our patients reported the development of a rash in the distribution of the nerves affected by Cryo. We initially treated this empirically as an infection, increasing our reported surgical site infection rate in the Cryo group. However, we started including the possibility of a self-resolving rash as part of our postoperative education once a pattern was observed. While 90% of our patients were discharged within 2 days of surgery, two patients, however, required a stay longer than 4 days due to chest tube placement for pneumothorax.

During each follow-up visit, our patients pre- and post-Cryo implementation were queried over neuropathic pain-related symptoms, and those who still complained of pain after a week of NSAID treatment were prescribed Gabapentin for management of neuropathic pain. Our two groups did not have a statistically significant difference in the occurrences of pain requiring Gabapentin prescription, suggesting that one minute per level Cryo, at the very least, is not associated with increased risk of neuropathic pain after SCOPE. This finding is in agreement with a recent report showing no significant increase in the incidence of neuropathic pain following Cryo [27].

Cryo, however, did not seem to decrease outpatient phone calls from patients and families to surgery clinic. These findings of high phone call burden are comparable to a recently published paper that also suggests higher levels of discomfort in Cryo patients than previously perceived [40].

In addition to improved pain control and decreasing resource utilization, we found that Cryo is associated with significantly lower total hospital costs. This finding is likely due to the significant decrease in the hospital LOS, and it parallels another recent report on cost of SCOPE with and without Cryo [13,23]. Rettig et al. found higher operating room costs but lower total costs associated with their Cryo group, demonstrating that prolonged hospital stay was the main driver for the cost in NoCryo group, while operating room costs tended to occupy a large proportion of cost for the Cryo group [23]. Our results in terms of decreasing opioid utilization and hospital length of stay using Cryo with a concomitant decrease in cost should lead to wider insurance coverage of the cryoablation procedure.

Our study has multiple limitations including the retrospective nature of the review. In addition, despite protocolizing pain management regimens, our protocols allowed for some freedom for individual patient and provider variation which may affect the total opioid utilization. We attempted to account for this variation by reporting medians which are known to be less affected by outliers than averages. Also, our study does not include patients who underwent more than one bar placement, this is due to change in surgeons' preference in favor of placing one bar over two bars over the past few years, even for patients with a higher Haller index. This practice pattern limits the generalizability of our results to patients undergoing SCOPE with more than one bar. Another limitation is that using retrospective chart review limited our ability to quantify the number of pills returned to clinic; however, recorded data suggested decreased outpatient opioid use in most patients. Lastly, due to the timeline of Cryo implementation, Cryo group had an overall shorter follow-up. However, all patients were followed for at least 90 days to account for complications.

5. Conclusion

Children undergoing surgical correction of pectus excavatum with one minute per level intercostal nerve cryoablation have a shorter hospital length of stay with far less need for systemic opioids compared to conventional analgesia. Pediatric surgeons performing correction of chest wall deformities should consider offering this technique. Further research into the optimal time of

cryoablation in terms of pain control and sensation recovery is needed.

Level of evidence

Level III.

Study type

Retrospective Comparative Study.

Previous communication

None.

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